

Notes: Caballero and Hammour (QJE, 1996)

On the Timing and Efficiency of Creative Destruction

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1 Big picture

- **Question.** How should an economy time job creation and job destruction when technological progress requires continual replacement of old production units with new ones?
- **Core mechanism.** Technological progress is embodied in new production units. Growth therefore requires *creative destruction*: old units must be scrapped and workers must be reallocated into new units.
- **Efficient timing.** In an efficient economy, creation and destruction are synchronized and concentrated in recessions because the opportunity cost of keeping workers unemployed for reallocation is lowest when current production is least valuable.
- **Market failure.** Incomplete contracting between capital and labor creates an appropriability problem. Once capital, training, organization, or search costs are sunk, workers or insiders can appropriate quasi-rents. This distorts hiring, destruction, wages, and unemployment.
- **Cyclical implication.** Appropriability decouples job creation and job destruction. Instead of recessions being periods of efficient joint reallocation, recessions can feature low creation, high destruction, and excessive unemployment.
- **Long-run implication.** Transactional difficulties can create technological sclerosis: renovation is too slow, old units survive too long, and productivity growth is inefficiently embodied too slowly in production.
- **Policy implication.** Production subsidies reduce unemployment but can worsen sclerosis. Creation subsidies accelerate reallocation. A suitable combination of production and creation incentives can restore efficiency.
- **Main lesson.** Creative destruction is not automatically efficient. The timing and pace of restructuring depend critically on contracting frictions in labor and capital markets.

2 Incremental contribution of the paper

2.1 Contribution relative to Schumpeterian growth theory

- Schumpeterian models emphasize that growth proceeds through innovation and replacement of old technologies.
- Caballero and Hammour add the macroeconomic timing dimension: when should the replacement of old production units happen over the business cycle?
- They show that an efficient economy tends to use recessions as periods of accelerated restructuring, because the opportunity cost of reallocation is low.

Interpretation: The paper does not simply say that growth destroys old jobs. It studies the efficient cyclical organization of creative destruction.

2.2 Contribution relative to business-cycle models

- Standard business-cycle models often treat recessions as periods of low factor utilization and low output.
- This paper interprets recessions as potentially useful times for reallocation.
- It shows why efficient recessions could be “cleansing” but also why actual recessions can be inefficient when contracting frictions decouple creation and destruction.

Interpretation: The paper gives a disciplined theory of cleansing recessions while rejecting the conclusion that laissez-faire recessions are necessarily efficient.

2.3 Contribution relative to search and matching models

- Search models emphasize unemployment as a matching friction.
- Caballero and Hammour reinterpret unemployment as a reallocation device: unemployment can temporarily release labor from old units and make workers available for new units.
- The paper then asks whether the unemployment generated by market equilibrium is too high, too low, or badly timed.

Interpretation: The paper's distinctive point is that the efficiency of unemployment depends on whether it supports productive reallocation or reflects distorted destruction without creation.

2.4 Contribution relative to contracting and hold-up theory

- Incomplete contracting and asset specificity generate hold-up problems in microeconomic theory.
- The paper embeds this appropriability problem in a dynamic general equilibrium model of growth, job flows, and business cycles.
- It shows that a micro-level contracting friction can generate macro-level sclerosis, excessive unemployment, and distorted cyclical job flows.

Interpretation: The incremental contribution is to connect hold-up in production-unit formation with aggregate labor-market dynamics and macroeconomic policy.

2.5 Contribution relative to policy debates

- A production subsidy can alleviate unemployment but can slow renovation.
- A creation subsidy can increase the pace of restructuring but may overshoot if used too strongly.
- The optimal policy uses both instruments to correct two different margins: creation costs and shadow wages.

Interpretation: The paper moves beyond a simple pro- or anti-recession view. It shows which margins must be subsidized or taxed to restore efficient creative destruction.

3 Model: a renovating economy

3.1 Production units and embodied technological progress

The economy contains production units, each of which combines one unit of capital, one unit of labor, and a unit of a nonproduced intermediate good. A unit created at time t embodies the frontier productivity $A(t)$, with

$$\dot{A}(t)/A(t) = \gamma > 0. \quad (1)$$

Each production unit is a job. New units are created at rate $H(t)$. Existing units fail exogenously at rate δ and are also scrapped endogenously when they are too old. Let $a(t)$ be the age of the oldest active unit. Aggregate employment and output are

$$E(t) = \int_0^{a(t)} H(t-a)e^{-\delta a} da, \quad (2)$$

$$Q(t) = \int_0^{a(t)} A(t-a)H(t-a)e^{-\delta a} da. \quad (3)$$

Interpretation: Employment is the stock of active production units. Output is the productivity-weighted stock of active units. Scrapping old units and creating new ones are therefore the key margins of growth.

3.2 Creation costs

Creating new units is costly. The total creation cost is

$$C(H, U)A(t) = [I(H) + S(H, U)H]A(t), \quad (4)$$

where $I(H)$ is investment, installation, training, and organizational cost, and $S(H, U)$ is the unit search cost. The paper often uses the functional form

$$I(H) = c_0 H + \frac{1}{2} c_1 H^2, \quad (5)$$

$$S(H, U) = c_2 (H/U)^{\eta/(1-\eta)}, \quad c_0, c_1, c_2 \geq 0, \quad 0 \leq \eta < 1. \quad (6)$$

Interpretation: The parameters c_1 and c_2 matter for cyclical dynamics. Convex investment costs smooth creation, while search costs make unemployment affect hiring costs and wage pressure.

3.3 Consumers, unemployment, and shocks

There is a continuum of workers of total mass L . Aggregate unemployment is

$$U(t) = L - E(t). \quad (7)$$

The price of the nonproduced intermediate good, normalized by frontier productivity, is $p(t)$. It is taken as an exogenous cyclical variable. A high $p(t)$ lowers the profitability of operating existing units because the intermediate input is more expensive.

Interpretation: Business cycles enter through variation in the marginal profitability of production. The model asks whether creation and destruction respond efficiently to this variation.

4 Efficient creative destruction

4.1 Planner's problem

A planner chooses creation $H(t)$ and destruction age $a(t)$ to maximize aggregate utility subject to resource constraints. The key margins are:

- the **destruction margin**: whether the oldest active production unit should be scrapped;
- the **creation margin**: whether new production units should be created now or later.

The efficient scrapping condition equates the productivity of the marginal old unit with the opportunity cost of keeping its labor and intermediate input employed:

$$A(t - a(t)) - [p(t) + C_U(t)]A(t) = 0. \quad (8)$$

The efficient creation condition equates the marginal creation cost with the discounted future net output stream from a newly created unit.

Interpretation: Efficiency depends on synchronizing creation and destruction. If the planner destroys a job, the released worker is most valuable when creation also rises to absorb that worker.

4.2 Why recessions are efficient reallocation times

When $p(t)$ is high or production profitability is low, the opportunity cost of unemployment is low. The efficient planner therefore concentrates both destruction and creation in recessions.

- Destruction rises because low-profit old units should be scrapped.
- Creation rises because unemployed labor is cheap and reallocation is socially valuable.
- Unemployment rises temporarily as a buffer stock that facilitates reallocation.

Interpretation: An efficient recession is not just a downturn. It is a period in which the productive structure is renovated.

5 Decentralized equilibrium and appropriability

5.1 Appropriability problem

Appropriability arises when capital investments, training, organizational costs, or search costs become partly specific to workers or groups of workers. After the investment is sunk, workers or insiders may bargain for part of the quasi-rents.

- Appropriability reduces the private return to creating a production unit.
- It makes wages less responsive to aggregate conditions.
- It distorts both hiring and scrapping decisions.

Interpretation: The problem is a macroeconomic form of hold-up. The private creator of a job does not receive the full social return from creating it.

5.2 Insider strength

The paper distinguishes between strong and weak insiders. Strong insiders extract too large a share of rents, which discourages creation and creates sclerosis. Weak insiders capture too little, which can lead to excessive creation relative to social value.

Interpretation: The same model can generate different inefficient regimes. The empirically important case emphasized in the paper is strong insiders: excessive unemployment plus slow technological renovation.

5.3 Creation-destruction decoupling

With appropriability, recessions no longer necessarily synchronize creation and destruction. The decentralized economy can have:

- high destruction because old units become unprofitable;
- low creation because private returns to new units are depressed;
- high unemployment because the released workers are not absorbed into new units.

Interpretation: This is the key business-cycle distortion. Efficient creative destruction requires creation and destruction to move together. Appropriability makes them move apart.

6 Sclerosis and steady-state distortions

6.1 Technological sclerosis

Technological sclerosis is the state in which old production units survive too long and the average age of the productive structure is too high. The economy renovates too slowly.

- Old units embody outdated technology.
- Creation of new frontier units is too slow.
- Productivity growth embodied in new units is not incorporated fast enough.

Interpretation: Sclerosis is a dynamic inefficiency. It is not merely unemployment at a point in time, but a persistent failure to replace old technologies with new ones.

6.2 Unemployment-sclerosis trade-off

In the strong-insider region, the economy suffers from both high unemployment and sclerosis. Policies that reduce unemployment by keeping old units alive may worsen sclerosis. Policies that accelerate creation can improve renovation but may affect unemployment ambiguously.

Interpretation: The policy problem has two dimensions. One instrument cannot generally fix both unemployment and the pace of restructuring.

7 Policy analysis

7.1 Production incentives

A production subsidy makes it more attractive to keep existing units operating. This can reduce unemployment but may delay destruction of obsolete units.

- Benefit: fewer workers are unemployed.
- Cost: old technologies remain in operation too long.
- In the strong-insider region, a small production subsidy is welfare improving because the unemployment distortion dominates locally.

Interpretation: The paper rejects a pure liquidationist view. Even if recessions cleanse, deliberately depressing production to force cleansing is not generally welfare improving.

7.2 Creation incentives

A creation subsidy directly stimulates formation of new production units. It raises the pace of reallocation and can reduce sclerosis.

- Benefit: new frontier technologies enter faster.
- Cost: if too large, it can produce excessive restructuring or hyperkinesis.
- In the strong-insider region, small creation subsidies are welfare improving.

Interpretation: Creation incentives are the natural tool for sclerosis because they target the margin that appropriability depresses.

7.3 Optimal dynamic policy

The paper shows that efficiency can be restored by combining creation and production incentives. The policy must correct both:

1. the effective creation-cost signal;
2. the shadow-wage signal at the destruction margin.

The optimal policies are more countercyclical when appropriability is more severe. During recessions, firms need stronger incentives because wages do not fall enough and unemployment is excessively high.

Interpretation: Optimal policy is not simply “subsidize jobs.” It must distinguish between keeping old jobs alive and creating new productive jobs.

8 Figures

8.1 Figures I and II: efficient cycles and job flows

Read:

- Figure I shows the efficient response to an aggregate shock. Creation and destruction rise together in downturns, and unemployment acts as a reallocation buffer.
- Figure II contrasts efficient joint smoothing with inefficient decoupling. With appropriability, creation falls while destruction rises.

Economic message: Efficient creative destruction synchronizes job creation and job destruction. Appropriability breaks this synchronization.

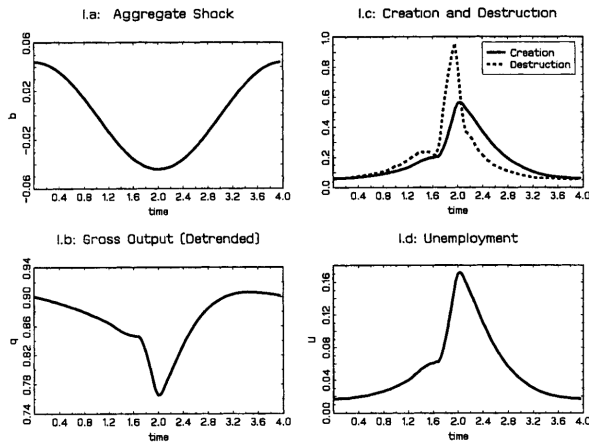


FIGURE I
Business Cycles in an Efficient Economy
Note that $c_0 > 0$, $c_1 = 0$, and $c_2 > 0$.

Figure I: efficient business cycles

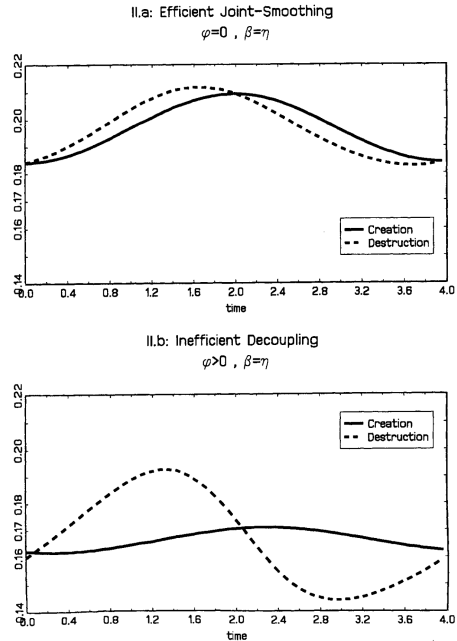


FIGURE II
Efficient and Inefficient Job Flows
Note that $c_0 > 0$, $c_1 > 0$, and $c_2 > 0$.

Figure II: efficient versus inefficient job flows

8.2 Figures III and IV: distorted volatility and inefficient business cycles

Read:

- Figure III shows that creation becomes insufficiently volatile and destruction excessively volatile when the economy is inefficient.
- Figure IV shows an inefficient business cycle in which creation and destruction no longer behave like coordinated reallocation margins.

Economic message: The key inefficiency is not simply more volatility, but the wrong composition of volatility: too much destruction without enough creation.

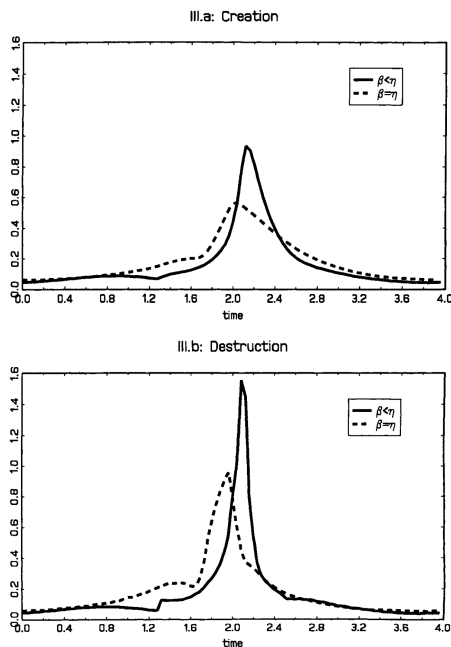


FIGURE III
Distorted Volatility
Note that $c_0 > 0$, $c_1 = 0$, and $c_2 > 0$.

Figure III: distorted volatility

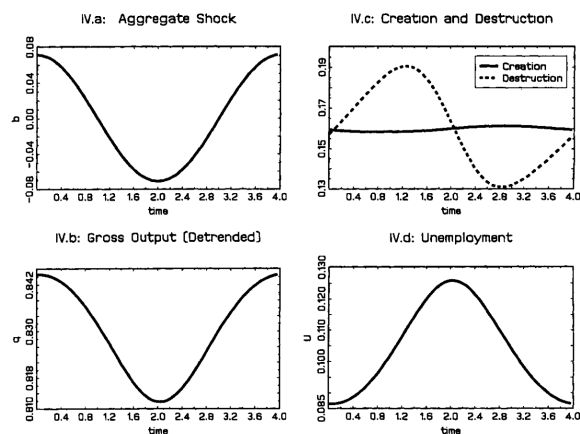


FIGURE IV
Business Cycles in an Inefficient Economy
Note that $c_0 > 0$, $c_1 > 0$, and $c_2 = 0$.

Figure IV: inefficient business cycles

8.3 Figures V and VI: Beveridge curves and insider regions

Read:

- Figure V compares Beveridge curves in efficient and inefficient economies.
- Figure VI divides parameter space into strong-insider and weak-insider regions.

Economic message: Unemployment-vacancy movements reveal how efficiently the labor market reallocates workers. Insider strength determines whether equilibrium is characterized by sclerosis or excessive renovation incentives.

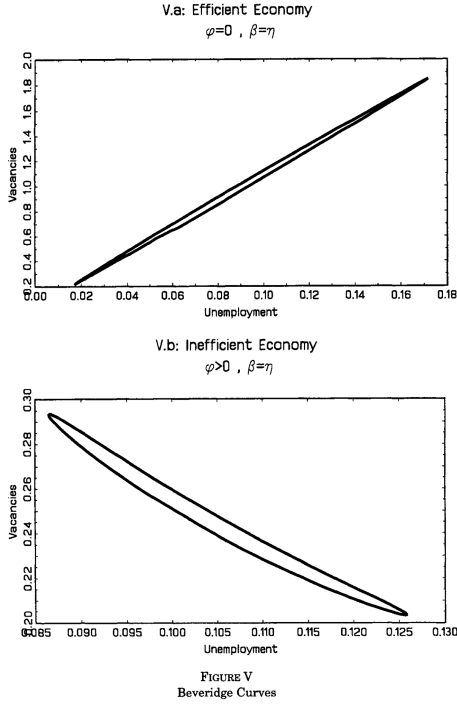


Figure V: Beveridge curves

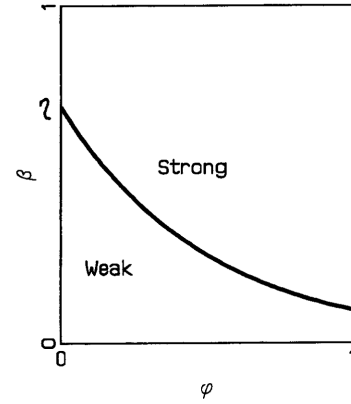


Figure VI: strong and weak insiders

8.4 Figures VII and VIII: steady-state distortions and policy margins

Read:

- Figure VII traces how steady-state unemployment and the age of the productive structure vary with insider bargaining/appropriability parameters.
- Figure VIII shows how production and creation incentives move the economy in unemployment-sclerosis space.

Economic message: Production and creation incentives operate on different margins. Production incentives mainly reduce unemployment, while creation incentives mainly reduce sclerosis.

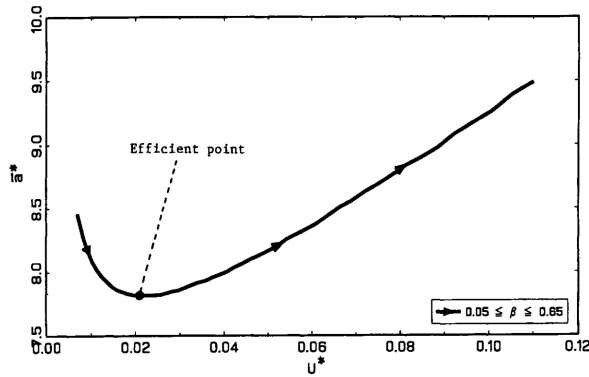


Figure VII: effects of increasing bargaining/appropriability parameter

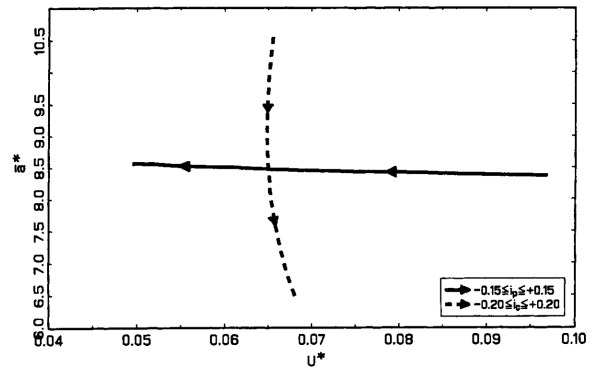


Figure VIII: production and creation incentives

8.5 Figure IX: optimal countercyclical policies

Read:

- Figure IX plots the business-cycle indicator and optimal paths of creation and production incentives.
- As appropriability worsens, optimal policies become more countercyclical.
- Recessions require stronger subsidies to creation and production because wages are too rigid and unemployment is excessive.

Economic message: Policy should lean against the inefficient decoupling of creation and destruction. In recessions, the policy system should encourage new job creation as well as support production.

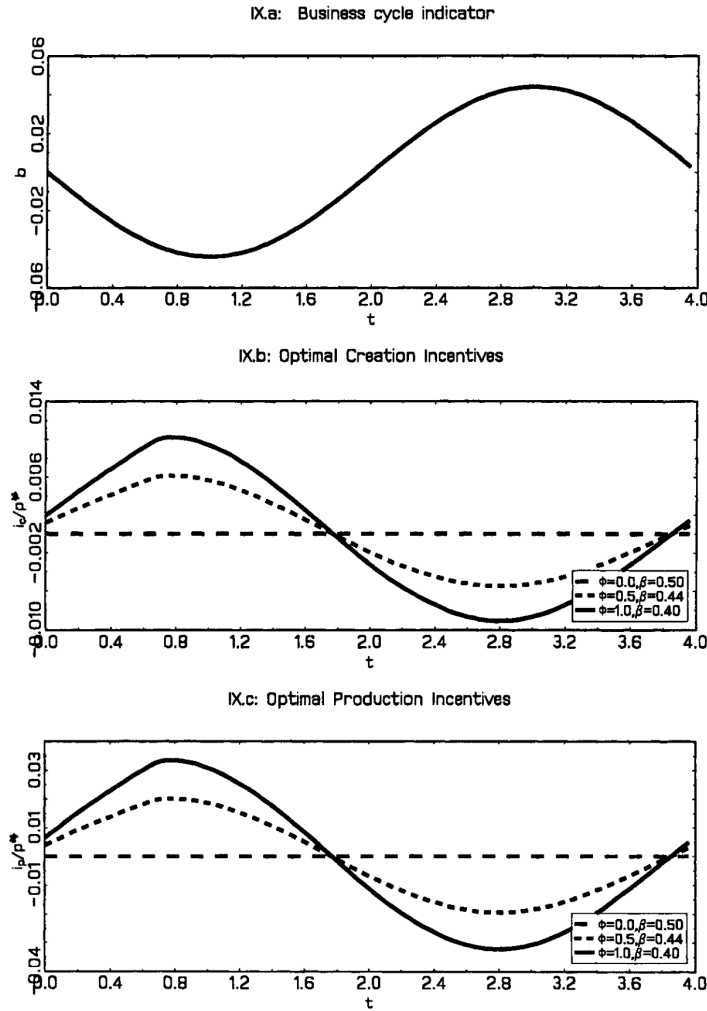


FIGURE IX
Optimal Countercyclical Policies
Note that $c_0 > 0$, $c_1 > 0$, and $c_2 > 0$.

9 Short summary

- Creative destruction requires continual creation of new production units and destruction of obsolete ones.

- In an efficient economy, reallocation is concentrated in recessions because the opportunity cost of unemployment is low.
- Efficient recessions combine high destruction with high creation.
- Incomplete contracting and appropriability distort this pattern.
- Appropriability decouples creation and destruction: recessions feature high destruction, low creation, and excessive unemployment.
- In the long run, contracting frictions can create technological sclerosis, with old technologies surviving too long.
- Production subsidies reduce unemployment but can worsen sclerosis.
- Creation subsidies accelerate renovation and reduce sclerosis.
- An optimal combination of production and creation incentives can restore efficiency.

10 Conclusion

10.1 Core conclusion

Caballero and Hammour show that creative destruction has an efficient timing: recessions should be periods of synchronized creation and destruction. However, labor-capital contracting frictions disrupt this timing and can turn recessions into periods of excessive destruction, insufficient creation, and high unemployment.

10.2 Economic intuition

Technological progress is embodied in new production units. Reallocation is therefore necessary for growth. But creating new units requires investments that become partly specific to workers. If insiders can appropriate the quasi-rents from those investments, private creation incentives fall. Old units may survive too long, and recessions may destroy jobs without generating enough new jobs.

10.3 Takeaway

The paper's main lesson is that the efficiency of creative destruction depends on institutions governing labor-capital contracting. Neither laissez-faire liquidation nor simple job preservation is generally optimal. Efficient policy must distinguish between support for production and incentives for creation, especially over the business cycle.